



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCTZ401

TERRAZZO WORKS

Perform Terrazzo Works

Competence

RQF Level: 4

Learning Hours



Credits: 8

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC4001- TVET Certificate 4 in Building Construction

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Purpose statement	This module describes the knowledge, skills, and attitude required to perform terrazzo works on building construction sites. It is intended to be pursued by learners at TVET level IV in building construction. At the end of this module, the learner will be able to perform terrazzo preliminary works, produce precast and cast in situ terrazzo, and install pre-casted terrazzo with respect to the standards.						
Leaning assumed to be in place	Concrete works, Tile works, Building construction drawing and safety at workplace.						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	• Group project and presentation	30%					
	• Individual project /Work	70%					
				Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Perform preliminary works	1.1. PPE is properly used according to the work to be done.
	1.2. Tools are appropriately selected in accordance to the work to be done.
	1.3. Equipment is appropriately selected and used in accordance to the work to be done.
	1.4. Storage of materials is properly prepared according to the standards.
	1.5. Materials are appropriately selected according to the work to be done.
	1.6. Working area is properly prepared according to the work to be done.
	1.7. Casting area is correctly prepared according to the design specifications.
2. Produce precast terrazzo tiles	2.1 . Types of terrazzo are correctly identified based on production methods.
	2.2 . Materials of terrazzo tiles are properly batched based on the design ratio.
	2.3 . Materials of terrazzo tiles are properly mixed based on the design ratio.
	2.4 . Mixed materials are correctly poured into the mold according to the designed shape.
	2.5 . Terrazzo tile top is correctly formed in accordance to the given design.

	2.6 . Terrazzo tile top is properly grinded and polished according to the designed finishing.
	2.7 . Terrazzo is conveniently cured according to the standards.
3. Install pre-casted terrazzo tile	3.1 Laying place is properly prepared in accordance to design.
	3.2 Mortar is correctly mixed according to the design ratio.
	3.3 Mortar is correctly applied to the surface area according to the standards.
	3.4 Pre-casted terrazzo tile is properly laid and levelled according to the given design.
	3.5 Grouting is properly done according to the standards.
	3.6 Pre-casted terrazzo tile is conveniently cured in accordance with the standards.
	3.7 Pre-casted terrazzo is accurately cleaned according to the standards.
4. Cast terrazzo on the floor	4.1 Floor/pavement is correctly prepared based on the work to be done.
	4.2 The floor is accurately set out according the designed pattern.
	4.3 Terrazzo materials are correctly batched according to the designed ratio.
	4.4 Terrazzo materials are correctly mixed in accordance to the designed ratio.
	4.5 Terrazzo mixed materials are properly poured to the floor in accordance to the standards.
	4.6 Poured terrazzo materials are adequately grinded and polished in accordance to the design.
5. Apply terrazzo on wall	5.1 Wall is properly cleaned and wet according to the standards.
	5.2 Wall is properly levelled according to the standards.
	5.3 Terrazzo materials are correctly batched according to the designed ratio.
	5.4 Terrazzo materials are correctly mixed in accordance to the mixing method.
	5.5 Terrazzo paste is properly applied to the surface based on the design.
	5.6 Terrazzo surface is properly cured according to the standards.
	5.7 Terrazzo surface is properly sealed according to the standards.
	5.8 Terrazzo surface is properly polished according to the standards.
6. Close terrazzo works	6.1 Equipment is properly cleaned according to the work.
	6.2 Tools are properly cleaned according to the work.
	6.3 Tools are adequately stored according to their types.

6.4 Equipment is adequately stored according to their category and status.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify terrazzo materials ✓ Identify terrazzo tools and equipment ✓ Differentiate different types of terrazzo and their uses ✓ Identify concrete mix for different areas of terrazzo application	✓ Handle and use Tools, Terrazzo equipment and material ✓ Read and interpret construction drawings and schedules ✓ Plan work and procedures of activities ✓ Mark out work areas ✓ Mix terrazzo ingredients ✓ Pour and cure terrazzo	✓ Demonstrate awareness of workplace hazards and associated control measures ✓ Demonstrate persistent, team working, flexibility, punctuality at work ✓ Demonstrate the ability to Work independently with minimal supervision ✓ Respect time assigned to every task ✓ Pay attention to details

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform preliminary works. 2. Produce precast terrazzo tiles. 3. Install pre-casted terrazzo tile. 4. Cast terrazzo on the floor. 5. Apply terrazzo on wall. 6. Close terrazzo works.
Learning outcome 1: Perform preliminary works	Learning hours: 5
Indicative content	

- **Identification of PPE used for terrazzo works**
 - Definition of PPE
 - Types of PPE
 - ✓ Helmet
 - ✓ Gloves
 - ✓ Goggles
 - ✓ Dust mask
 - ✓ Overall
 - ✓ Safety shoes(boots)
 - ✓ Ear protector
 - Uses of PPE
 - Maintenance of PPE
- **Select tools and equipment for terrazzo works**
 - Classification of tools and equipment for terrazzo works
 - ✓ Setting out tools/equipment
 - ✓ Cutting tools/machines
 - ✓ Mixing tools/equipment
 - ✓ Batching tools/equipment
 - ✓ Smoothing tools/equipment
 - ✓ Cleaning tools/equipment
 - ✓ Setting out tools/equipment
 - ✓ Placing tools/equipment
 - ✓ Transporting equipment
- **Preparation of tools and equipment for terrazzo materials**
- **Storage of terrazzo materials**
 - Types of storage
 - Requirements of good storage
- **Selection of terrazzo materials**
 - Introduction to terrazzo
 - Identification of terrazzo materials
 - ✓ Marble chips
 - ✓ Quartz
 - ✓ Granite
 - ✓ Glass
 - ✓ Matrix
 - ✓ Admixtures
 - Factors for selecting terrazzo materials
- **Preparation of working area**
 - Assessment of working area
 - Cleaning of the working area

➤ Watering the surface area • Preparation of casting area ➤ Set out the casting area ✓ wall/column ✓ base flooring	
Resources required for the learning outcome	
Equipment	PPEs, Wheelbarrow, Trucks, Air compressor, Dumpy level
Materials	Building lines, Timber, Water, Fuel, Nails, Pegs, Warning tape, Marker pen, Marble chips, Quartz, Granite, Glass, Matrix, Admixtures, cement, epoxy
Tools	Wire Brushes, Straight edge, Spade, Shovel, Hoe, Pickaxe, Machete Spirit level, Hammers (Club and claw), Brooms, Metallic/Plastic pans, Metallic/Plastic buckets, Jerrycan, Hose pipe
Facilitation techniques	• Demonstration • Brainstorming • Simulation • Think pair square share
Formative assessment methods	• Written assessment • Matching questions • Multiple choice • Short questions • Open ended questions

Learning outcome 2: Produce precast terrazzo tiles	Learning hours: 25
Indicative content	
• Identification of terrazzo types ➤ Type of terrazzo ✓ Epoxy terrazzo ✓ Cement terrazzo • Batching of terrazzo materials ➤ Methods of terrazzo materials batching ✓ Weight batching ✓ Volume batching ➤ Procedure of fabricate wooden gauge box • Mixing of terrazzo materials ➤ Terrazzo mix ratio	

- Methods of mixing terrazzo materials
 - ✓ Hand mixing
 - ✓ Machine mixings
- Procedures of mixing terrazzo materials
- **Procedures of pouring terrazzo materials paste into the mould**
- **Method of the decorative pattern on the top of the Terrazzo**
 - ✓ Brass angle placement
 - ✓ Webbing placement
- **Grinding the top of terrazzo tile**
 - Methods of grinding
 - ✓ Wet method
 - ✓ Dry method
 - Functions of grinding
 - Importance of grinding
 - Process of grinding
- **Rolling and beating the terrazzo**
 - Importance of rolling and beating
 - Techniques of rolling and beating
- **Steps of polish and sealing the top of terrazzo tile**
- **Cure the terrazzo tile**

Resources required for the indicative content

Equipment	PPEs, Wheelbarrow, Trucks, beam balance, Gauge box, Belt conveyor, Chute, Power floating machine, grinding machine, Beam balance, polishing machine, rolling machine
Materials	Building lines, Timber, Water, Fuel, Warning tape, Marble chips, Quartz, Granite, Glass, Matrices , Admixtures,
Tools	Straight edge, Spade, Shovel, Hoe, Trowels, Spirit level, Hammers (Club and claw) Squares, Metallic/Plastic pans, Metallic/Plastic buckets, Floats, Jerrycan ,Hose pipe Plumb bob, Building lines, Bucket/Jerrycan
Facilitation techniques	• Trainer guided • Brainstorming • Simulation • Demonstration with video aid
Formative assessment methods	• Performance assessment • Written assessment • Multiple choice • Matching questions • Sentence completion

- Open ended questions

Learning outcome 3: Install pre-casted terrazzo tile

Learning hours: 10

Indicative content

- **Preparation of laying place**
 - Cleaning
 - Setting out
 - Applying water
- **Mix mortar**
 - Types of mortar
 - Method of mixing mortar
 - ✓ Hand mixing
 - ✓ Machine mixing
 - Mixing mortar tips
- **Application of mortar**
- **Laying and leveling the pre-casted terrazzo tile**
 - Select and mark the starting point/intersection
 - Place the first terrazzo tile at the marked intersection
 - Place a tile spacer in each corner of the tile
 - Applying thin set mortar and keep adding terrazzo tiles and spacers
- **Grouting the pre-casted terrazzo tile**
 - Types of grouting materials
 - Importance of Grouting
 - Grouting procedures
 - ✓ Allow the thin set mortar to set hard
 - ✓ Remove the tiles spacers
 - ✓ Apply the grout
 - ✓ Remove the excess grout from joints
- **Curing the pre-casted terrazzo tile**
 - Methods of curing
 - Period of curing
- **Cleaning the pre-casted terrazzo**
 - Wash all surfaces
 - Properties of liquid cleaner

Resources required for the indicative content

Equipment	PPEs, Wheelbarrow, Mortar mixer, Trucks, water pump, Hose pipe, Bucket, water tank, Air compressor, watering pipe, Polishing machine
Materials	Building lines, Timber, Spacers, Chalk, Marker pen, Water, Fuel, Warning tape,

	Cement, Epoxy, Terrazzo tile
Tools	Straight edge, Spade, Shovel, Trowels, Spirit level, Hammers (Club and claw) Squares, Metallic/Plastic pans, Metallic/Plastic buckets, Floats, Jerrycan, Hose pipe Plumb bob, Building lines, Bucket
Facilitation techniques	• Demonstration • Trainer guided • Brainstorming • Simulation • Think pair square share
Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions

Learning outcome 4: Cast terrazzo on the floor	Learning hours: 20
Indicative content	
• Preparation of the Floor/pavement ➤ Check the Floor/pavement defects ➤ Repair defects or cracks ➤ Check moisture ➤ Level the surface • Setting out the floor ➤ Method of set out ➤ Procedure of set out • Batching of terrazzo materials ➤ Methods of batching ✓ Weight batching ✓ Volume batching • Mixing of terrazzo materials ➤ Apply mix ratio	

- Methods of mixing terrazzo materials
 - ✓ Hand mixing
 - ✓ Machine mixing
- Procedures for mixing terrazzo materials
- **Pouring to the floor terrazzo materials mixture**
- **Rolling and beating terrazzo materials**
 - Importance of rolling and beating
 - Procedures of rolling and beating
- **Grinding terrazzo materials mixture**
 - Factor affecting the quality of workpiece surfaces
- **Polishing the poured terrazzo materials mixture**
 - Methods of polishing
 - ✓ Recrystallization,
 - ✓ Use of polishing powder,
 - ✓ Diamond polishing

Resources required for the indicative content

Equipment	PPEs, Wheelbarrow, Trucks, Concrete pump, Concrete mixer, Belt conveyor, Chute, Power floating machine, grinding machine, Beam balance, polishing machine
Materials	Building lines, Timber, Spacers, Chalk, Water, Fuel, Warning tape, Cement, Epoxy, strip divider, Marble, chips, granite, quartz, Glass,
Tools	Straight edge, Spade, Shovel, Hoe, Trowels, Spirit level, Hammers (Club and claw), Squares, Metallic/Plastic pans, Metallic/Plastic buckets, Floats, Jerrycan, Hose pipe , Plumb bob, Building lines, Bucket
Facilitation techniques	• Trainer guided • Demonstration • Simulation • Brainstorming • Practical exercises
Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions

- Open ended questions

Learning outcome 5: Apply terrazzo on the wall

Learning hours: 15

Indicative content

- Assessment of the wall surface
- Cleaning and wetting the wall
- Procedures of setting out the surface area
- Steps used for applying terrazzo paste on the wall surface
 - Levelling the wall surface
 - Batching and mixing terrazzo materials
 - Spreading terrazzo paste
 - Screeding the surface area
 - Curing terrazzo wall surface
 - Seal terrazzo wall surface
 - Polish terrazzo wall surface

Resources required for the indicative content

Equipment	PPEs, Wheelbarrow, Trucks, Concrete pump, Concrete mixer, Dumpy level, Belt conveyor, Chute, Power floating machine, grinding machine, Beam balance, polishing machine, Compacting roller
Materials	Building lines, Timber, Spacers, Chalk, Water, Fuel, Warning tape, Cement, Epoxy, Terrazzo tile
Tools	Straight edge, Spade, Shovel, Hoe, Trowels, Spirit level, Hammers (Club and claw), Squares, Metallic/Plastic pans, Metallic/Plastic buckets, Floats, Hose pipe Plumb bob, Building lines, Bucket/Jerrycan
Facilitation techniques	• Trainer guided • Simulation • Demonstration • Brainstorming

Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions
Learning outcome 6: Close terrazzo works	Learning hours: 5
Indicative content	
• Clean tools and equipment ➤ Purposes of cleaning tools and equipment ➤ Methods of cleaning tools and equipment • Clean the workplace ➤ The importance of a clean workplace ➤ Methods of Cleaning the workplace • Store tools and equipment ➤ Importance of proper storage of tools and equipment ➤ Points to follow in storing tools and equipment ➤ Process of storing tools and equipment	
Resources required for the indicative content	
Equipment	PPEs, Wheelbarrow, Air compressor, water pipe, Tipper truck
Materials	Water, Fuel, oil, soap,
Tools	Bucket, broom, brush, spade
Facilitation techniques	• Trainer guided • Brainstorming • Demonstration • Think pair square share
Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions

Integrated/Summative assessment (For specific module)

Integrated situation

SWM has a tender to apply terrazzo in their building office located at KIYOVU in NYARUGENGE district. NJJ Construction company ltd won a tender to perform those activities, the dimensions of the floor are 4m*2m, and height of the wall is 2.80 m. As a terrazzo technician, you are hired to perform the following tasks:

1. Cast terrazzo on the floor of building office.
2. Apply the pre-casted terrazzo on the four walls of building office.
3. Apply the pre-casted terrazzo on the skirting of 20cm height

All materials are available at NJJ Construction company ltd stores. The mixing ratio of 1:2 must be respected, all task will be performed within 8 hours

Resources

Tools	Tape measure, building square, Spades, Trowel, Spirit level, Clow Hammer, chisel, Brush, Straight edge, Float, Metallic/Plastic buckets, Floats, Jerrycan, Horse pipe level, Plumb bob, Straight edge.
Equipment	PPEs, Wheelbarrow, Air compressor, Trucks, Concrete pump, Concrete mixer, Belt conveyor, Chute, Power floating machine, grinding machine, polishing machine, Mortar mixer
Materials/ Consumables	Building lines, Timber, Chalk, Water, Fuel, Warning tape, Cement, Epoxy, Concrete, mortar, fine aggregate, Coarse aggregate, water, Marble chips, Terrazzo tile, Strips divider, Granite, Glass, Quartz, Matrix, admixture

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Perform preliminary works (15%)	1.1. PPE is properly used according to the work to be done	Helmet is used			0.5
		Overall is used			0.5
		Safety boot is used			0.5
	1.2. Tools are appropriately selected in accordance to the work to be done	Setting out tools are selected			0.5
		Batching and mixing tools are selected			0.5
		Cutting and smoothing tools are selected			0.5
		Cleaning tools are selected			0.5
	1.3. Equipment is appropriately selected and used in accordance to the work to be done	Batching and mixing equipment are selected			0.5
		Cutting and smoothing equipment are selected			0.5
		Transport and placing equipment are selected			0.5
		Cleaning equipment are selected			0.5
	1.4. Materials are appropriately selected according to the work to be done	Marble is selected			0.5
		Quartz is selected			0.5
		Granite is selected			0.5

		Glass is selected			0.5
		Matrix is selected			0.5
		Admixture is selected			0.5
		Aggregate is selected			0.5
		Cement is selected			0.5
		Water is selected			0.5
		strip divider is selected			0.5
	1.5. Working area is properly prepared according to the work to be done	Working area is cleaned			0.5
		Working area is watered			0.5
		Working area is levelled			1
	1.6. Casting area is correctly prepared according to the design specifications	Defects are checked			0.5
		Defects are repaired			1
		Casting area is set out			1
Learning outcome 3: Install pre-casted terrazzo tile (25%)	3.1. Mortar is correctly applied to the surface area according to the standards	Fresh mortar is applied to the wall surface			2
		Fresh mortar is applied to the skirting surface			2
	3.2. Pre-casted terrazzo tile is properly laid and levelled according to the given design	Starting point is selected			3
		Starting point is marked			3
		First terrazzo tile is placed on the marked point			3
		Strip dividers are placed			2
	3.3. Grouting is properly done according to the standards	Strip dividers are removed			2
		All joints are grouted			2
		Excess grouts are removed			2
	3.4. Pre-casted terrazzo tile is conveniently cured in accordance to the standards	Installed terrazzo tile is cured			2
	3.5. Pre-casted terrazzo is accurately cleaned according to the standards	Installed terrazzo tile is cleaned			2
Learning outcome 4: Cast terrazzo on the floor (25%)	4.1. Floor/pavement is correctly prepared based on the work to be done	Floor defects are checked			1
		Floor defects are repaired			1
		Moisture contents is checked			1
		Floor surface is levelled			1
	4.2. The floor is accurately set out	Strip dividers are arranged			2

	according the designed pattern	The floor dimension is checked			1
	4.3. Terrazzo materials are correctly batched according to the designed ratio	Terrazzo materials are batched			1
		Mix ratio is respected			2
	4.4. Terrazzo materials are correctly mixed in accordance to the designed ratio	Terrazzo paste is homogeneous			1
		Terrazzo paste is workable			1
		No impurities observed in terrazzo paste			1
		Procedures of mixing is respected			2
	4.5. Terrazzo mixed materials are properly poured to the floor in accordance to the standards	Fresh terrazzo paste is poured to the floor			2
		Fresh terrazzo paste is rolled			2
		The poured terrazzo paste is beaten.			2
	4.6. Poured terrazzo materials are adequately grinded and polished in accordance to the design	Surface area is smooth			1
		Surface area is brilliant			1
		Surface area is levelled			1
		Surface area is uniform			1
Learning outcome 5: Apply terrazzo on wall (25%)	5.1. Wall is properly cleaned and wet according to the standards	Wall is cleaned			1
		Wall is watered			1
	5.2. Wall is properly levelled according to the standards	Wall is levelled			2
	5.3. Terrazzo materials are correctly batched according to the designed ratio	Quantity of matrices are batched			1
		Quantity of granite is batched			1
		Quantity of quartz is batched			1
		Quantity of marble chip is batched			1
		Quantity of admixture is batched			1
		Quantity of water is batched			1
	5.4. Terrazzo materials are correctly mixed in accordance to the mixing method.	Terrazzo paste is homogeneous			1
		Terrazzo paste is workable			1
		No impurities observed in terrazzo paste			1
		Procedures of mixing is respected			2
	5.5. Terrazzo paste is properly applied to the surface based on the design	Dividing strips are put in place			2
		Terrazzo paste is applied to the wall surface			2
	5.6. Terrazzo surface is properly cured according to the standards	Terrazzo surface is cured			1

	5.7. Terrazzo surface is properly sealed according to the standards	Terrazzo surface is sealed			1
		Terrazzo surface is shining			1
	5.8. Terrazzo surface is properly polished according to the standards	Appearance of the surface wall is observed			2
		Terrazzo surface is smooth			1
Learning outcome 6: Close terrazzo works (10%)	6.1. Tools and equipment are properly cleaned according to the work	Setting out tools/equipment are cleaned			1
		Batching tools/equipment are cleaned			1
		Mixing tools/equipment are cleaned			1
		Cutting tools /equipment are cleaned			1
		Smoothing tools/equipment are cleaned			1
	6.2. Tools and equipment are adequately stored according to their types.	Setting out tools/equipment are stored			1
		Batching tools/equipment are stored			1
		Mixing tools/equipment are stored			1
		Cutting tools /equipment are stored			1
		Smoothing tools/equipment are stored			1
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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